AS THE MIDDLE EAST CONFLICT REVEALS HIGHER LEVELS OF DANGER THROUGH CCTV SYSTEMS, THE MODI GOVERNMENT HAS PUT DOWN STRICT RULES TO RESTRICT AND REGULATE THEIR OPERATION

BY **PRADIP R. SAGAR**

For years, security experts have cried themselves hoarse over the rapid proliferation of surveillance cameras. Adversaries hacking internet-connected CCTVs (Closed Circuit Televisions) to gather caches of sensitive information is bad enough, but the ongoing war in the Middle East has revealed the more sinister uses such hacking can enable. Cameras—at traffic junctions, buildings and city streets—are being repurposed for target tracking and precision strikes. Advances in artificial intelligence (AI) allow militaries to rapidly analyse vast caches of footage, enabling precise identification of targets. Israel hacked Tehran's surveillance camera grid to track—every face and vehicle was tabulated over days to create a mosaic of daily routines—and then assassinate Iranian supreme leader Ayatollah Ali Khamenei and other senior officials on February 28.

In India, the threat from CCTVs was brought home in late March, when police in Ghaziabad, Uttar Pradesh, dismantled a Pakistan-linked espionage ring that did more than hack existing cameras. Instead, operatives installed solar-powered CCTV units at sensitive locations in Delhi-NCR. These devices transmitted live footage to Pakistan's Inter Services Intelligence (ISI) handlers across the border. Investigators revealed that the network was tasked to expand the covert network in other Indian cities.

Indeed, there are over three million CCTVs with weak technical features installed across the country—around half of them in government premises, including military compounds and high-security infrastructure—a massive security risk. Ninety per cent are made by Chinese firms like Hikvision and Dahua, which means there is but one beneficiary of alleged electronic espionage—Beijing.

With India's apprehension turning into reality in the Middle East, New Delhi has finally made a decisive move. The government has banned CCTV cameras and associated hardware from Chinese manufacturers. From April 1, new certification requirements under the Union ministry of electronics and information technology's (MeitY) Standardisation Testing and Quality Certification (STQC) framework come into force, mandating that all CCTV products undergo testing and approval before they can be sold or deployed. The rules require devices to have tamper-proof enclosures and strong malware detection encryption so that they are free of vulnerabilities that could lead to remote access. The new rules stem from the requirements first issued by MeitY in April 2024. Manufacturers were given a two-year period to adapt, and over 500 CCTV models have already been certified and some installed.

SECURING VITAL ASSETS

At a review meeting chaired by Prime Minister Narendra Modi in February, serious concerns were also raised about alleged Chinese attempts to snoop on India's national telecommunications infrastructure. The government is now establishing a mechanism to safeguard the country's telecom backbone, with quarterly coordination meetings mandated between the Department of Telecommunications, the Ministry of Defence and intelligence agencies.

The perceived threat to vital national infrastructure stems from a large cyber-espionage campaign in 2021 by Chinese state-linked groups. The operation targeted at least seven State Load Dispatch Centres (SLDCs)—facilities responsible for electricity dispatch and grid control—located close to the Line of Actual Control (LAC) with China in Ladakh. The attackers likely compromised Internet Protocol (IP) cameras commonly used in CCTV networks, along with internet-connected Digital Video Recorder (DVR) systems. The objective, it was assessed, was to gather intelligence on India's power infrastructure. Two separate Chinese hacking attempts on

INDIA'S NEW SURVEILLANCE POLICY

From April 1, any internet-connected CCTV camera manufactured, imported or sold in India must be certified by the MeitY's Standardisation Testing and Quality Certification (STQC) protocol. This includes mandatory evaluation of:

- **Country of origin of key parts like firmware, SoC**
- **Backdoor/ hidden access vulnerabilities**
- **Data encryption standards**
- **Firmware integrity & security controls**
- **Chinese brands/ parts refused certification, effectively barred**



"Internet-connected CCTV cameras can be used for surveillance and denial of service attacks. Security is a must and the government move is a good step"

LT GEN. RAJESH PANT (RETD)
Former National Cybersecurity Coordinator